CS3 Case Study Rubric: Sentiment Analysis of Restaurant Reviews
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Formatting	● Clear organization with labeled sections ● Concise and readable writing ● Code runs without errors ● All required components are included ● Includes clear explanation of results in written form
Task Overview	**Goal:** ● Build a model that classifies restaurant reviews as positive, neutral, or negative using only the text of the reviews ● Evaluate model performance, compare the two models, and explain key findings in a short written summary
Data Preparation	**Goal:** Understand and verify the dataset used for modeling ● Use the provided cleaned dataset ● Confirm that sentiment labels are correctly assigned ● Check for missing or invalid values in the dataset **Meets Expectations:** ● Dataset is correctly loaded ● Student verifies that the data is usable ● Student demonstrates understanding of how the data is structured
Feature Creation	**Goal:** Convert text into numerical features ● Use TF-IDF to represent review text ● Apply consistent feature settings across the dataset ○ (e.g., same vocabulary and parameters for all data) **Meets Expectations:** ● TF-IDF is implemented correctly ● Features are appropriate for model input
Modeling	**Goal**: Train a model to classify sentiment ● Use classification models (Logistic Regression and Naive Bayes) ● Split data into training and testing sets ● Address class imbalance (e.g., through class weighting) **Meets Expectations:** ● Model is trained correctly ● Training/testing split is applied ● Model choice is appropriate
Evaluation	**Goal:** Assess model performance ● Report accuracy on test data ● Examine performance across sentiment classes using confusion matrices and classification reports ● Identify strengths and weaknesses of the model

	Meets Expectations: ● Accuracy is calculated correctly ● Performance is interpreted (not just reported) ● Limitations are acknowledged ● Student correctly interprets confusion matrices and classification reports
Interpretation & Insights	**Goal:** Explain what the model reveals ● Discuss patterns in predictions ● Compare the performance of the two models ● Identify challenges (e.g., difficulty classifying neutral sentiment) ● Explain impact of class imbalance ● Reflect on why the model performs as it does **Meets Expectations:** ● Insights go beyond surface-level ● Clear explanation of results ● Demonstrates understanding of model behavior
Success Criteria	● A strong model achieves ~70% accuracy or higher ● The full pipeline (data understanding → features → model → evaluation) is completed ● The student provides a clear explanation of results and limitations ● The student demonstrates understanding of both model performance and its limitations